

A MASH 111 All-Digital Delta-Sigma Modulator with Hybrid 2nd-Order Shaped Dithering for Fractional-N Frequency Synthesizers

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Abstract—This work proposes a hybrid 2nd-order shaped dithering method for the MASH 111 all-digital delta-sigma modulator. The proposed method injects a hybrid 1st-order shaped dither signal into the 2nd and 3rd stages of the MASH 111 delta-sigma modulator, enables it to avoid any tonal spur in its output quantization noise power spectrum, and at the same time alleviates the influence of the dither signal in the low frequency zone by hybridly shaping it to the 2nd order.

Keywords—Dither; MASH; delta-sigma modulator; fractional-N frequency synthesizer.

I. INTRODUCTION

Nowadays, the delta-sigma fractional-N frequency synthesizer is widely adopted in wireless transceiver chips. The block diagram of a typical delta-sigma fractional-N frequency synthesizer is shown in Fig.1, from which it is easy to find that the all-digital delta-sigma modulator is one of its key building blocks. Usually, the delta-sigma modulator quantizes the fraction part of the frequency control word (FCW) into integer and at the same time shapes the concomitant quantization noise to high frequency zone. The output (Y) of the delta-sigma modulator is a pseudo-random number sequence, which is used to change the instantaneous division factor of the multi-modulus frequency divider, so as to achieve the desired long-term average fractional frequency division.

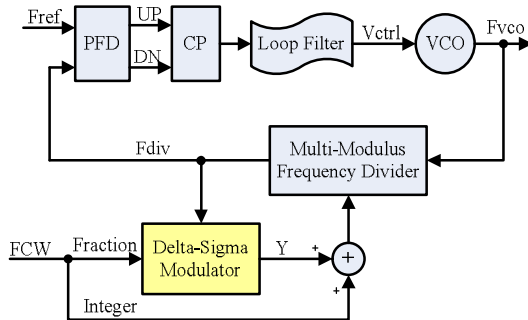


Figure 1. A typical delta-sigma fractional-N frequency synthesizer.

By now, many different delta-sigma modulator architectures have been proposed for fractional frequency synthesis applications, among which the 3rd-order multistage-noise-

shaping architecture (MASH 111 DSM) is the most popular one [1]. The schematic of a classical pipelined n-bit MASH 111 DSM is shown in Fig.2. It consists of a cascade of three n-bit 1st-order DSM's and a delta path. The popularity of MASH 111 DSM results from its following merits: (1) its circuit is quite simple, only a few adders and registers are needed; (2) it is unconditionally stable for any input in the range of [0, 1), which avoids any output frequency gap resulting from modulator instability; (3) its output quantization noise is only 3rd-order shaped, so a relatively simple and easy 4th-order phase-locked loop is enough to filter out the quantization noise that is pushed into the high frequency zone; (4) if its inner quantization noises (E_1 , E_2 and E_3) are random enough to be approximated as white noise, then its output quantization noise power spectrum will be smooth and spur-free, which means that the synthesizer will not suffer from fractional spurs.

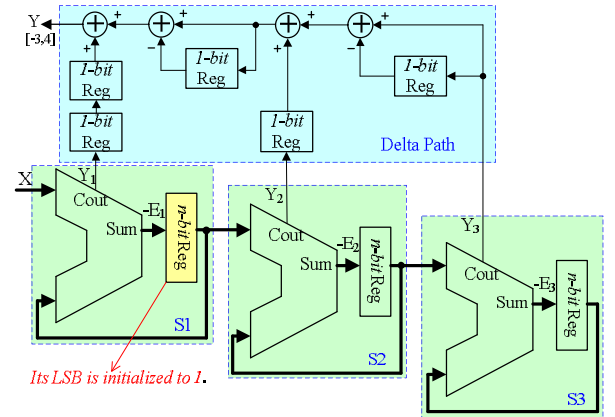


Figure 2. The MASH 111 delta-sigma modulator proposed in [2].

To achieve all the above-mentioned merits of MASH 111 DSM, the requirement that all its inner quantization noises should be approximately as random as white noise must be first met. In fact, it is the most important prerequisite to the practicability of MASH 111 DSM. However, as MASH 111 DSM is a finite-state machine, when its input X equals some special values like 0, 0.25, 0.5 and 0.75 and when it starts from the all-zero initial condition, its state space will be cycle-limited, which makes its inner quantization noise sequences repeat periodically, resulting in strong tonal spurs in its output quantization noise power spectrum. A solution to this problem

is to reset the least-significant bit (LSB) of the n-bit error register in the first stage (S1) to 1 whenever the input X changes. This solution was first empirically found by the designers of [2] and was then mathematically verified by the work of [3, 4]. It was shown that after adopting this “resetting LSB to 1” solution, an N-bit MASH 111 DSM gains a guaranteed output sequence length of 2^{n+1} and its output quantization noise power spectrum is totally spur-free.

Except for the “resetting LSB to 1” method, dithering is another effective method to randomize the inner quantization noises. In the recent work of [5], the output bit stream of an m-bit linear feedback shift register (LFSR) is used to replace the LSB’s of the inputs to the 2nd and 3rd stages of a MASH 111 DSM, so as to achieve the goal of randomizing the quantization noises sufficiently without any influence on the DC input signal X. Fig.3 shows a MASH 111 DSM adopting the “LFSR dithering” method of [5], in which the LFSR dither signal is inputted to the carry-in ports (Cin) of the 2nd and 3rd stages for simplifying the design process. The z-domain linear mathematical model of this MASH 111 DSM can be deduced as follows:

$$Y_1(z) = X(z) + (1-z^{-1})E_1(z) \quad (1)$$

$$Y_2(z) = -z^{-1}E_1(z) + (1-z^{-1})E_2(z) + X_d(z)/2^n \quad (2)$$

$$Y_3(z) = -z^{-1}E_2(z) + (1-z^{-1})E_3(z) + X_d(z)/2^n \quad (3)$$

$$\Rightarrow Y(z) = z^{-2}Y_1(z) + (1-z^{-1})[z^{-1}Y_2(z) + (1-z^{-1})Y_3(z)] \\ = z^{-2}X(z) + (1-z^{-1})^3E_3(z) + (1-z^{-1})X_d(z)/2^n \quad (4)$$

Equation (4) shows that for the MASH 111 DSM of Fig.3, its output consists of three parts: the input signal with two clock periods delay $z^{-2}X(z)$, the 3rd-order shaped quantization noise of the 3rd stage $(1-z^{-1})^3E_3(z)$ and the 1st-order shaped dither signal $(1-z^{-1})X_d(z)/2^n$. The theoretical proof and experimental results of [5] both demonstrate that its proposed dithering method is quite effective in avoiding tonal spurs and achieves a smooth 3rd-order shaped output quantization noise power spectrum. However, as (4) indicates, the added dither signal is only 1st-order shaped when it passes through the MASH 111 DSM, which increases the noise power level in the low frequency zone. As a result, the close-in phase noise performance of the synthesizer will potentially be degraded.

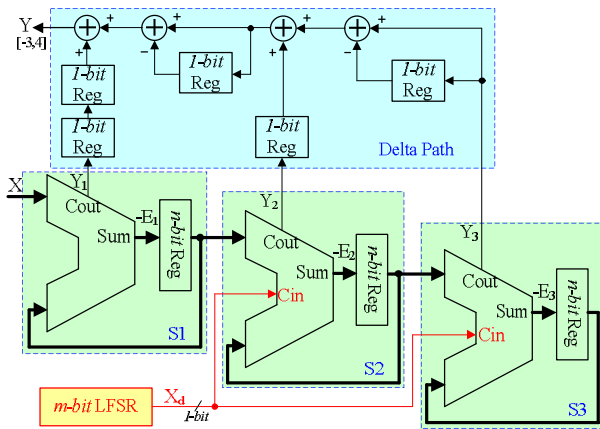


Figure 3. The MASH 111 delta-sigma modulator proposed in [5].

In this work, we propose a hybrid 2nd-order shaped dithering method for MASH 111 DSM. The proposed method not only keeps the outstanding randomization effect of the “LFSR dithering” method, but also reduces the influence of the added dither signal in the low frequency zone. In Section II the proposed method will be explained in detail. In Section III experiment results of MASH 111 DSM’s using the three different quantization noise randomization methods will be demonstrated and their performance will be compared. Finally a conclusion is given in Section IV.

II. HYBRID 2ND-ORDER SHAPED DITHERING

The schematic of a MASH 111 DSM with our proposed hybrid 2nd-order shaped dithering method is shown in Fig.4. Comparing with the “LFSR dithering” method of [5], we just add a few simple digital circuits to process the output bit stream of the m-bit LFSR before it is fed to the carry-in ports of the 2nd and 3rd DSM stages. The hybrid dither signal we finally get is:

$$X_h(z) = (1-z^{-1})X_d(z) + 1/4 + (1-z^{-1})E_d(z) \\ = (1-z^{-1})[X_d(z) + E_d(z)] + 1/4 \quad (5)$$

Replacing the $X_d(z)$ of (4) with (5), we can get the output of our proposed MASH 111 DSM as follows:

$$Y(z) = z^{-2}X(z) + (1-z^{-1})^3E_3(z) + (1-z^{-1})X_h(z)/2^n \\ = z^{-2}X(z) + (1-z^{-1})^3E_3(z) \\ + (1-z^{-1})^2[X_d(z) + E_d(z)]/2^n + (1-z^{-1})^*1/2^{n+2} \quad (6)$$

Equation (5) and (6) actually reveal our key design idea. Firstly we use a simple differentiator to get the output of the m-bit LFSR 1st-order shaped. As the differentiation result is a 2-bit signed number, which does not match the 1-bit unsigned carry-in inputs of the 2nd and 3rd DSM stages, we then add 1 to it so as to eliminate its sign, and finally we use a 2-bit 1st-order DSM to compress the 2-bit unsigned data sequence into the usable 1-bit 1st-order shaped dither signal. Obviously, the dither signal we generated contains not only the LFSR output part $X_d(z)$ but also an extra quantization noise part $E_d(z)$. That’s the first reason why the proposed dithering method is called “hybrid”. On the other hand, after passing through the delta path of the MASH 111 DSM, the injected 1st-order shaped dither signal is differentiated once again, which makes it finally present a 2nd-order shaped power spectrum in the output of the MASH 111 DSM. The fact that the two differentiation operations happen at different segments of the proposed DSM is referred as the second reason for its being called “hybrid”. Another interesting characteristic of our proposed DSM is that there seems to exist an extra component $(1-z^{-1})^*1/2^{n+2}$ in its output, which results from the 1 we added for sign elimination. However, as $1-z^{-1}=0$ at DC and the added 1 is constant and only contains power at DC, the component $(1-z^{-1})^*1/2^{n+2}$ actually equals 0, which means it does not influence the final output of the proposed DSM.

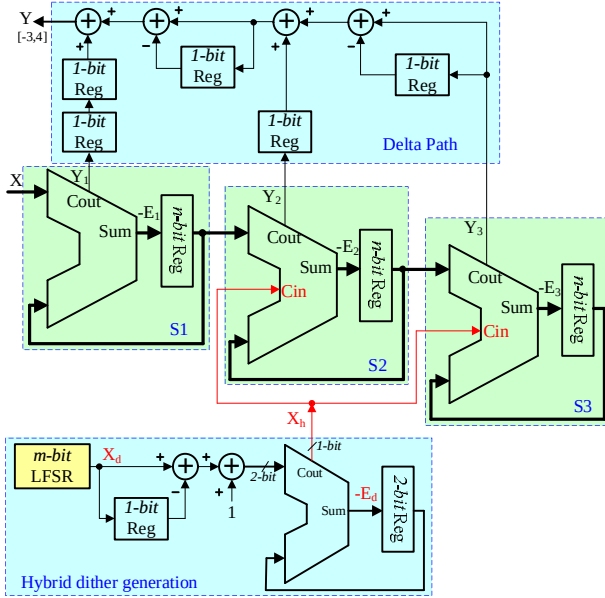


Figure 4. The MASH 111 delta-sigma modulator proposed by this work.

III. PERFORMANCE EVALUATION

In order to compare the performance of different DSM quantization noise randomization methods, we implement three 17-bit MASH 111 DSM's using Verilog HDL at the RTL level. The first DSM adopts the "resetting to 1" method of [2], the second one adopts the "LFSR dithering" method of [5] and the third one adopts our proposed hybrid 2nd-order shaped dithering method. The LFSR's in the last two DSM's are both 7-bit wide. Thorough logic simulations are carried out for various input conditions, and 2²⁰ samples of each DSM output are collected and then analyzed with Discrete Fourier Transformation, assuming a sampling frequency of 20MHz. The resultant discrete output quantization noise power spectra are then used for performance comparison.

Fig.5 and Fig.6 are two examples of the enormous experiment results. They are picked out for demonstration because their inputs are respectively 0 and 0.5, which are both spur-prone input conditions for conventional MASH 111 DSM. Apparently, the three randomization methods all work well enough to avoid any tonal spur in their output quantization noise power spectra. And the "LFSR dithering" method and our proposed hybrid dithering method are better than the "resetting to 1" method in the high frequency zone because their output quantization noise power are about 6dB lower, which implies their output sequences are about 4 times as long as that of the "resetting to 1" method [4]. In the low frequency zone, the influences of the added dither signals are obvious. As we have expected, the "LFSR dithering" method presents a 1st-order shaped low-frequency dither noise and our proposed hybrid dithering method presents a 2nd-order shaped low-frequency dither noise. Moreover, it can be seen that the low-frequency noise of our proposed hybrid dithering method is much lower than that of the "LFSR dithering" method, which proves the superiority of our method.

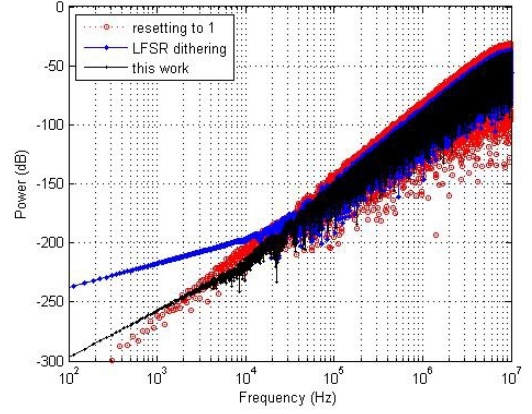


Figure 5. The output quantization noise power spectra of the three MASH 111 delta-sigma modulators for X=0.

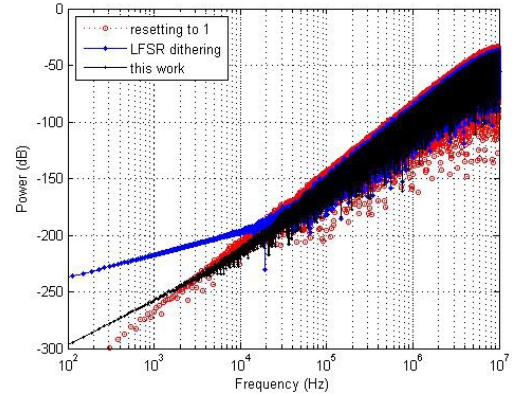


Figure 6. The output quantization noise power spectra of the three MASH 111 delta-sigma modulators for X=0.5.

IV. CONCLUSION

This paper has demonstrated that a hybrid 1st-order shaped dither signal can be injected into the 2nd and 3rd stages of the MASH 111 delta-sigma modulator, which enables the modulator to avoid any tonal spur in its output quantization noise power spectrum, and at the same time the influence of the dither signal in the low frequency zone is alleviated by hybridly shaping it to the 2nd order.

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